S3 Supplementary QC and Robustness

Support boundaries and L4 certificate audit

Article type: Supplement

Short title: S3 QC ds004511

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S3.1 Coverage

Chart-Jacobian alpha is finite on 134/134 recordings. Lag-1 one-step quantities are computed on 121/134 recordings; history-conditioned L2 is computed on 115/134. Lag-2 is absent. These are support facts, not successful imputation targets.

S3.2 L4 certificate failures

Subject	Task	Gate	Failure
sub-S200121	Rest	F	post_dtype_psd_failure
sub-S200214	GG	F	post_dtype_psd_failure
sub-S210203	CC	F	post_dtype_psd_failure

Table 1: All missing finite reach_a_q certificates. One record falls in each task and each subject is distinct.

S3.3 Degenerate readout

Of 131 finite reach_d_eff values, 109 (83.2%) lie within $10^{-}6$ of 1; the median and minimum are 1 and the maximum is 1.31377. This rank-degenerate ellipsoid summary cannot support a capacity-dimension contrast. Apparent Wilcoxon significance on residual numerical variation is therefore method-limited and excluded from the article's substantive claims.

S3.4 Next QC gate

The bounded audit is complete. The 13 recordings comprise 12 Rest and 1 CC (0 GG), all on a regular 4-s grid with 120–260 lag-1 pairs. Each fails the existing out-of-fold fidelity gate (median(rel-MSE fold 1, fold 2) < 0.9) with one_step_fit_not_better_than_baseline; reproduced fold scores agree within 1.338e-07. Diagnostic threshold sweeps (not production changes) identify 126/134 at 0.95, 128/134 at 1.0, and 131/134 at 1.5. This identifies a reproducible support boundary, not a reason to fill absent lags or reopen the estimator.

S3.5 Fixed-duration chart-alpha sensitivity

All 134 recordings contain at least 100 exported chart-Jacobian alpha windows. Re-aggregation over the first 400 s preserves all three contrast directions: GG minus Rest, median difference 0.01450, $d_z = 1.119$, $n = 44$, $q = 2.76 \times 10^{-}8$; CC minus Rest, 0.00593, $d_z = 0.394$, $n = 45$, $q = 0.0124$; and GG minus CC, 0.00995, $d_z = 0.648$, $n = 44$, $q = 7.21 \times 10^{-}5$. Here q is BH over the three fixed-prefix alpha contrasts. This is a sampling-support sensitivity result, not a new measurement family or an assertion that early and full-recording dynamics are identical.

S3.6 Duration boundary for L4

Recording durations are non-overlapping by task: Rest 480–596 s, CC 904–1324 s, and GG 2396–4652 s. The export contains no per-window L4 series, so a balanced-prefix L4 analysis is unavailable without recomputing the estimator. Audit-only within-task associations are positive in GG for reach_a_q (Spearman $\rho = 0.348$, $p = 0.0224$) and reach_v_norm ($\rho = 0.353$, $p = 0.0204$). These uncorrected correlations neither prove nor remove duration dependence. Large L4 task contrasts remain method-limited until L4 is recomputed under a predeclared balanced-window contract.

S3.7 Fixed-duration L1 sensitivity

All four frozen L1 series and exported validity masks support the same 100-window prefix in all 134 recordings. The 12 metric-by-contrast rows were corrected as one sensitivity family. Effects generally attenuated relative to the full recordings; 11/12 median-difference signs agreed. The sole sign change was the negligible and non-significant CC-minus-Rest `diffusion_effective_dimension` row (full $d_z = 0.022$; fixed $d_z = -0.081$, fixed $q = 0.732$).

Contrast	L1 metric	Full d_z	400-s d_z	400-s q
GG – Rest	`diffusion_a_hat_frob`	0.520	0.445	0.00836
GG – Rest	`diffusion_total`	0.551	0.453	0.00836
GG – Rest	`diffusion_effective_dimension`	0.146	0.0667	0.759
GG – Rest	`chart_drift_b_hat_norm`	1.82	1.44	2.02e-9
CC – Rest	`diffusion_a_hat_frob`	0.166	0.272	0.219
CC – Rest	`diffusion_total`	0.155	0.266	0.304
CC – Rest	`diffusion_effective_dimension`	0.0218	−0.0811	0.732
CC – Rest	`chart_drift_b_hat_norm`	0.505	0.536	0.00586
GG – CC	`diffusion_a_hat_frob`	0.332	0.133	0.489
GG – CC	`diffusion_total`	0.433	0.162	0.392
GG – CC	`diffusion_effective_dimension`	0.182	0.193	0.380
GG – CC	`chart_drift_b_hat_norm`	1.43	0.947	1.37e-6

Table 2: Full-recording and deterministic 400-s-prefix L1 effect sizes. The displayed prefix q is BH over the 12 L1 sensitivity rows; full-recording q values belong to the separate 45-test task family and are not compared numerically here.

In the separate four-test fixed-prefix Friedman family, `chart_drift_b_hat_norm` remained informative ($\chi^2 = 36.05$, $q = 5.96 \times 10^{-}8$). The diffusion-norm omnibus rows attenuated to $q = 0.0564$ (`diffusion_a_hat_frob`) and $q = 0.0649$ (`diffusion_total`); effective dimension remained uninformative ($q = 0.761$). Thus the drift-norm pattern is robust to equalized sampling support, while the broader diffusion-family claim is weaker and contrast-specific.

The standardized chart-versus-L3 interaction uses 121 recordings with both production-supported estimators and complete subject contrasts of $n = 32$, $n = 32$, and $n = 43$ for GG–Rest, CC–Rest, and GG–CC. Its positive GG–Rest and GG–CC rows survive the separate three-test exploratory BH family, but the estimand depends on the declared pooled median/MAD normalization. It is therefore a common-support robustness and estimator-difference analysis, not a raw-unit comparison or a duration test.

QC verdict

PARTIAL PASS. Chart-alpha and L1 drift-norm directions survive the deterministic 400-s sensitivity, while several diffusion results attenuate and L4 duration dependence remains unresolved. Lag-1 support is partial; lag-2 is unavailable; `reach_d_eff` is method-limited. No claim of a complete one-step, diffusion-invariant, reachability, or dimension analysis is licensed.